

Impact of Personality on Technology Adoption: An Empirical Model

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An innovator's personality along with the perceived attributes of an innovation predicts the rate of diffusion. The current study focuses on the personality factors that determine the likelihood of adoption of a technological innovation. To that end, the study distinguishes between global innovativeness and context-specific innovativeness. An information processing model was tested where technological innovativeness was purported to be indirectly influenced by an individual's global innovativeness, through its impact on communication and media use behaviors. The structural model was tested on two separate technology clusters, and partial support was found for linking sophistication in information search, and prior technology ownership to technological innovativeness.

The diffusion of innovations paradigm provides explanations for when and how a new idea, practice, or technique is accepted, rejected, or reevaluated over time in a given society (Rogers, 1995). Though the theory covers all aspects of the diffusion process, its strength lies in its ability to structure and predict the factors leading to adoption decisions. According to Rogers (1995), the decision to adopt an innovation is predicted, in part, by the perceived attributes of an innovation, and the personality of the potential innovator. In recent years, a great deal of research attention has focused on the impact of perceived attributes on adoption (e.g., the technology acceptance model (TAM); Davis, Bagozzi, & Warshaw, 1989); however, the treatment of personality has been inconsistent especially within the communication sciences. One reason for this is the nature of the personality construct, and difficulties in consistently measuring its associated dimensions (Rogers, 1995). Often, researchers use differing strategies based on different conceptualizations, which result in nongeneralizable findings. For example, Kang (2002) measures adoptive innovativeness among cable subscribers using length of time as a digital cable subscriber as the measure; while Wei (2001) measures the same con-

struct among cellular phone non-adopters by measuring their ownership of an arbitrary cluster of telecommunication devices. Such differences in approach and measurement strategies have resulted in an inability to evaluate the reliability and validity of findings, which hampers theoretical and scientific development.

Against this background, the study focuses on the impact of personality and the antecedent determinants of innovativeness. It examines different measurement strategies and distinguishes between the various conceptualizations of innovativeness. Next, a research model is presented and tested within the context of technology adoption. In the first section of this article the theoretical premise is presented, followed by the research model and the hypotheses. Then, the methods and measures of variables are given in the second section. In the third section the analysis strategy, followed by the measurement model, the evaluation of the structural model, and the tests of the hypotheses are covered. This is followed by the discussion and the conclusions in the fourth section.

Theoretical Framework

Measures of Innovativeness

Rogers (1995) defines innovativeness as the degree to which an individual is relatively earlier in adopting an innovation than other members of a system. Based on this definition, a number of researchers take a measure of the time since introduction of an innovation as an indicator of innovativeness or use time-of-adoption to assign individuals to adopter categories based on some arbitrary categorization (Goldsmith & Hofacker, 1991). This strategy is especially popular among communication scholars when assigning individuals to adopter categories. For example, Leung and Wei (1999) used this strategy to categorize all those consumers who had not yet adopted cellular phones in Hong Kong by 1998 as "laggards." Likewise, when identifying adopter profiles within digital cable subscribers, Kang (2002) used market penetration at the time of study and assumed that all the consumers who had subscribed to digital cable by 1999 must be early adopters. Using this strategy to segment adopters has been criticized since time of adoption

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is a temporal concept that bears no isomorphic relationship with the construct “innovativeness” (Midgley & Dowling, 1978). In addition, this strategy does not effectively predict future rate of adoption and diffusion, since it requires the adoptive decisions to have already taken place.

Another strategy proposed by Midgley and Dowling (1978) is the use of a cross-sectional method wherein an individual's ownership of a particular list of products is compared to ascertain his degree of innate innovativeness. Difficulties in determining what constitutes an innovation, choosing the appropriate products within a category, and choosing the products that are complementary or supplementary to the innovation, reduces the reliability and generalizability of findings using this approach (Goldsmith & Hofacker, 1991).

The final approach is to directly identify innovators using self-report measures. There are two conceptually distinct dimensions of innovativeness that are often measured—global innovativeness and context-specific innovativeness. The former is a personality dimension that cuts across the span of human behavior, while the latter refers to innovative attitudes and behaviors within a certain category (Flynn & Goldsmith, 1993). Hence, measures for both these constructs are conceptually and empirically distinct (Foxall & Szmigin, 1999) and should not be substituted for each other.

Global innovativeness is defined as the degree to which an individual makes innovative decisions independently of the communicated experience of others (Midgley & Dowling, 1978). That is, it is an individual's predisposition to behave in a given way regardless of the stimuli that activates the behavior (Foxall & Szmigin, 1999). In this conceptualization, innovativeness is viewed as an enduring personality trait possessed to a greater or lesser degree by all individuals. It is believed to be a continuous variable normally distributed within a population and generalizable across products (Hirschman, 1980).

This concept is closer to Rogers's conceptualization of innovativeness. According to Rogers (1995), innovators have an obsession with “venturesomeness” and risk, which leads to a proclivity to adopt. This penchant for risk requires substantial financial resources, a high tolerance for uncertainty, and an ability to understand and apply complex technical knowledge (Rogers, 1995). Personality researchers contend that all generalized personality traits fall within five dimensions, known as the five-factor model (Buss, 1996) of personality. The five dimensions—Surgency, Agreeableness, Conscientiousness, Emotional Stability, and Openness to Experience, are enduring, with most people varying between the extremes (Goldberg, 1981, 1992; Hofstee, De Raad, & John, 1990). Global innovativeness can be subsumed within the openness-to-experience dimension, since the propensity to be innovative would require a predisposition to be open to new experiences. This dimension manifests a host of behaviors, including meeting new people, and seeking out and accepting new information about innovations. Hence, innovators tend to have more social participation, are more cosmopolite, have greater exposure to mass media, and seek

more information about an innovation (Rogers, 1995) than later adopters.

One self-report measure that has proved robust in predicting global innovativeness is Kirton's Adoption-Innovation Inventory (KAI) (Kirton, 1976). Kirton contends that everyone can be located on a continuum ranging from adaptive to innovative, and uses 32 general statements to ascertain the respondent's cognitive style. Kirton's Adoption-Innovation Inventory correlates with all of the cognitive-personality variables known to be associated with consumer innovation (Foxall, 1995). Details on the psychometric properties of KAI can be found in Kirton (1976).

On the other hand, context-specific innovativeness is defined as the tendency to learn about and adopt innovations within a specific domain of interest (Goldsmith & Hofacker, 1991). This concept views innovativeness as a personality state, rather than a global tendency to behave in a certain way. Measures of context-specific innovativeness assume that the propensity to innovate is a behavioral response to a specific context, which is for the most part determined by an individual's interest, experience, exposure, and knowledge of a product category. Personality plays a minimal role, and as such there is little or no innovativeness across domains or product categories (Gatignon & Robertson, 1985). Hence, “this construct mediates both conceptually and empirically the relationship between the generalized personality trait, innate innovativeness, and specific innovative behaviors” (Midgley & Dowling, 1978, p. 238).

One reliable measure of context-specific innovativeness, Goldsmith and Hofacker's (1991) Domain-Specific Innovativeness Scale, consists of six highly particular statements, which can be adapted to any domain of interest. These statements indirectly measure the respondent's involvement, knowledge, and experience within a particular category. Details on the psychometric properties of the scale can be found in Goldsmith and Hofacker (1991), and Flynn and Goldsmith (1993).

The Research Model

Based on this theoretical premise, the research model (Figure 1) attempts to integrate global innovativeness and domain-specific innovativeness to test the antecedents of technology adoption. Technology products were chosen as it was felt that this domain would benefit from an understanding of the antecedents of innovativeness, and due to ease in measurement using student samples. College students also tend to be interested, experienced, knowledgeable, and more likely to purchase a technology product, which adds to the reliability of findings. In addition, since socioeconomic levels tend to impact the expression of innovativeness, the student sample provides a level of control on this factor.

Rogers (1995) defines technology as “a design for instrumental action that reduces the uncertainty in the cause-effect relationships involved in achieving a desired outcome” (p. 12). According to Rogers, unlike most consumer products, the decision to adopt or reject technological

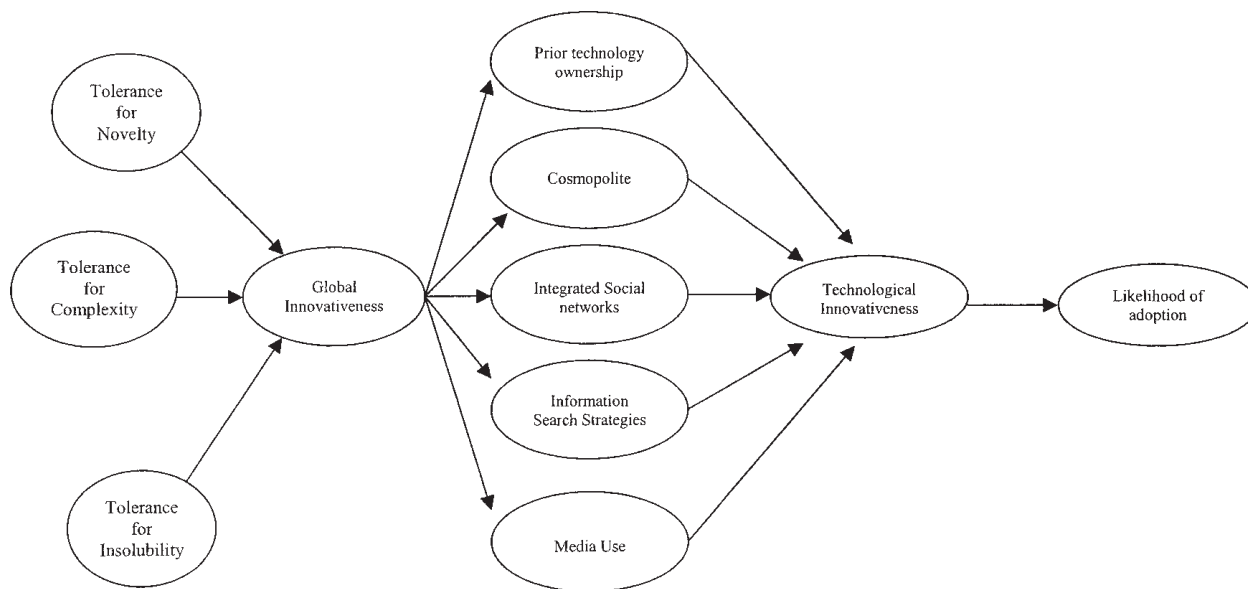


FIG. 1. Research model.

innovations involves higher levels of uncertainty since the expected consequences of their adoption are not always clear to the potential adopter. Technological products create uncertainty about their expected consequences, as well as present an opportunity to reduce uncertainty through its application and use (Rogers, 1995). Hence, the decision to adopt or reject a technological innovation is essentially an uncertainty reduction process. Information seeking is aimed at reducing uncertainty about the innovation to a tolerable level, in order to decide on a course of action (accept or reject the innovation).

Given high levels of uncertainty in technological adoption, innovators and early adopters would need to possess an innate personality trait that supports high levels of ambiguity in decision-making (Rogers, 1995). Scholars differentiate between three types of ambiguities: novelty, complexity, and insolubility. Novelty is experienced in a completely new situation in which there are no familiar cues; complexity is experienced in a situation in which there are a great number of cues; and insolubility is experienced in a contradictory situation in which the cues suggest different and often conflicting structures (Budner, 1962). For any new product, novelty translates to the lack of precedent in users and demonstration of actual usage, which tends to create high levels of uncertainty for the potential adopter. Hence, the tolerance for this type of uncertainty should be innately high to support innovativeness. In addition, most new technologies seldom represent a superior alternative to the previous product or service it replaces (Rogers, 1995). For example, the benefits of using MSN Explorer over AOL's web browser services are not obvious to an early adopter. Also, most technological innovations range in complexity, and require some degree of effort in developing a competency for its use. The perceived complexity of an innovation is hence considered a key attribute in predicting its diffusion. Therefore, innovators for technological products would need to possess a high tolerance for

complexity, and the willingness to devote effort in learning or coping with a new innovation. Lastly, competitive advertising often results in conflicting cues to a potential adopter. This is exemplified in the case of telecommunications products such as digital cable and direct broadcast satellite (DBS), where the ability to discern the veracity of conflicting advertising claims is usually not available until the adoption decision is made. In addition, since adopters often enter into service and subscription contracts such as for cable television and mobile telephony, the uncertainty involved and thereby tolerance for insolubility would need to be higher. Given the high levels of uncertainty in technology adoption, the research model places tolerance for ambiguity, complexity, and insolubility as a scope condition impacting innate innovativeness, and suggests the following hypothesis:

Hypothesis 1: Global innovativeness is positively related to tolerance for ambiguity, complexity, and insolubility.

Rogers (1995) presents a set of generalizations about the communication behavior of innovators. Based on this, the research model posits that global innovativeness mediates tolerance for uncertainty, and impacts communication and media use behaviors. Based on this, the following sets of hypotheses are proposed:

Hypothesis 2: Global innovativeness is positively related with ownership of technology products, social integration, "cosmopolitaness," information search behavior, and mass media exposure and use.

Hypothesis 3: Global innovativeness mediates the relationship between tolerance for ambiguity, complexity, and insolubility, and the above exposure variables.

Next, innovativeness within the domain of technology, i.e., technological innovativeness is posited as being indirectly

influenced by personality, and directly influenced by exposure. The research model hypothesizes an information-processing role of exposure on shaping behavior, and posits a direct influence of exposure on technological innovativeness.

Hypothesis 4: Ownership of technology products, social integration, cosmopolitanism, information search behavior, and mass media exposure and use is directly related to technological innovativeness.

Last, technological innovativeness is expected to mediate the relationship between media exposure, and predict the potential purchase of new technology products. The following hypothesis is proposed:

Hypothesis 5: Technological innovativeness mediates the relationship between media use, and predict the likelihood to adopt a new technology product.

Method and Measures

To test the antecedents of innovativeness, it was first necessary to ascertain the typical products that constitute the domain of technology. Technological products tend to be unlike most consumer products, in that they usually have a combination of hardware and software elements (Rogers, 1995), and the boundaries of the product categories are not easily defined. For example, one could consider cellular phones, PDAs, and personal computers, as three distinct product categories or as part of a complementary technology cluster. When considered separately, each product in itself would constitute a separate domain, i.e., domain of cellular phones would be distinct from that of PDAs, which would be distinct from personal computers. Innovators within each domain would be distinct and separate. However, empirical evidence seems to suggest that the ownership of technological products tend to stimulate the use of other functionally similar products (Atkin & LaRose, 1994; LaRose & Atkin, 1988). For example, cellular phone ownership and frequency of use is predicted by computer ownership (Vishwanath & Goldhaber, 2003). Hence, to ascertain the products that constitute a technology cluster, an initial list of 16 new technological innovations were drawn by three doctoral students. The attempt was made to include a broad range of products that spanned telecommunications, imaging, and compression technologies. Next, a questionnaire measuring likelihood of adoption, and current ownership was administered in an introductory communication class at the State University of New York at Buffalo.

The data collection yielded 131 usable responses, which were subjected to a principle component factor analysis. A varimax rotation yielded three factors that cumulatively explained 57% of the variation in the data. Factor 1 comprised PDAs, wireless text enabled PDAs, cellular phones with built in cameras, LCD flat panel monitors, satellite radios, and text-enabled cellular phones. Factor 2 comprised high speed Internet subscription, premium cable subscription,

DVD players, digital cameras, digital video cameras, video game consoles, and MP3 players. Factor 3 consisted of only two products that accounted for less than 5% of the variance, and was subsequently dropped. The final questionnaire was developed with specific questions eliciting technological innovativeness within two domains extracted in factors 1 and 2. This was administered to a separate undergraduate class, which yielded the data for the study. The final data collection yielded 260 usable responses.

Measures

All the constructs in the study are measured using multiple, fully anchored, 5-point Likert type scales ranging from *strongly disagree* to *strongly agree*, or *very likely*, to *not at all likely*. Whenever possible, initial scales were taken from previously validated measures in the literature, and reworded to relate to context of the study.

Likelihood to adopt technology. Respondents were first screened for current ownership of a list of technology products, and then probed on their likelihood to adopt technology products from within the list of 16 products. Based on the two extracted factors, the first domain of adoption consisted of six, and the second domain consisted of seven products.

Technological innovativeness. Technological innovativeness was measured within each technology domain using Goldsmith and Hofacker's (1991) six-item Domain-Specific Innovativeness scale. This scale is versatile and allows for the inclusion of product category names within each item. For example, Goldsmith and Hofacker (1991), and Flynn and Goldsmith (1993) used the scale to test for rock music innovativeness by adding the term "rock music" to each of the six items. Following this, the current study created items that included the names of all the products that potentially constituted a technology cluster. Screeners and written instructions ensured that only the products that the respondent did not already possess were considered when answering these questions.

Prior ownership of technology. Following Leung and Wei (1999), and Vishwanath and Goldhaber (2003), respondents were asked if they owned or subscribed to a cluster of 18 functionally similar technology products. Each adoption or subscription was dummy coded (1 = *yes*, 0 = *no*), and subsequently added up to yield a composite.

Cosmopolite. Respondents' degree of "cosmopolitanism" was measured using four items that measured the number of times they traveled and intended to travel to a destination apart from their hometown within the United States, and outside the United States. These variables measure the extent to which the respondent is oriented outside his or her social system.

Social networks. To elicit integration of social networks, respondents were asked to write the names of three of their closest friends, and answer three sets of questions about each of them. Respondents used a scale of 1 for *yes* and 0 for *no*, and recorded whether each friend owned technology products, whether the respondent had spoken to them in the recent past, and whether these friends speak to one another.

Information search. A new six-item scale probing sophistication in information search was developed based on Kellermann and Reynolds' (1990) six-item scale measuring information seeking during initial interaction, and Beatty and Smith's (1987) two-item information search scale.

Media use. Two items measured the number of minutes spent surfing the Internet, and spent sending, receiving, reading, and writing e-mail on the previous day. Respondents were also asked the minutes of television viewed in three separate parts in the previous day.

Global innovativeness. Innate innovativeness was measured using Kirton's 32-item Adoption-Innovation Inventory (1976). Each item probes the individual's proclivity to adapt and innovate, and has a theoretical range of scores from 32 to 160.

Tolerance for novelty, complexity, and insolubility. Tolerance for uncertainty was measured using the 16-item intolerance for uncertainty scale developed by Budner (1962). Each item uses a Likert-type scale to measure a person's intolerance for ambiguity, complexity, and insolubility.

Results

Analysis Strategy

First, the measurement model was assessed for its reliability and construct validity. The sample size for the study ($N = 260$) is large, and exceeds the minimum of 150 recommended by Anderson and Gerbing (1988). Hence, the research model was estimated using maximum likelihood (ML). ML is a large sample estimator that is especially robust to deviations from normality when the sample size is large (Hu, Bentler, & Kano, 1992). Next, the research model was evaluated for its fit, and the hypothesized relationships were examined.

The Measurement Model

The test of the measurement model includes estimation of the construct validity and reliability of the measures. Construct validity was evaluated by conducting a confirmatory factor analysis on Budner's (1962) intolerance for uncertainty scale, Kirton's Adoption-Innovation inventory, and Goldsmith and Hofacker's (1991) six-item Domain-Specific

Innovativeness scale. Scale reliabilities were measured using Cronbach's alpha. Because of the large number of variables in the design, the confirmatory factor analysis loadings were used to create the composite of observed variables, which loaded on their respective latent constructs. This latent composite approach accounts for unreliability by extracting measurement error from the latent constructs used in the structural model (Holbert & Stephenson, 2002).

For Budner's (1962) intolerance for uncertainty scale, most items did not closely meet the standard acceptable norm for Cronbach's alpha of 0.80. Items measuring intolerance for novelty reported 0.69, intolerance for complexity reported 0.58, and intolerance for insolubility reported 0.56. Similar low reliabilities were reported by Budner (1962) during initial scale construction. He attributes low reliability for his scale first, to the use of alpha rather than split half coefficients; second, to the freedom of the scale from artifacts such as acquiescence and social desirability which tends to maximize reliability estimates; and third, to the complex multidimensional nature of uncertainty itself which "increases the probability of individuals exhibiting unique patterning's of the component elements" (p. 35).

Scores for Kirton's Adoption-Innovation inventory can range from 32 to 160. The responses from 260 subjects ranged from 53 to 148, with a mean score of 101.44 ($SD = 14.33$), which was very close to the theoretical mean of 96 reported by Kirton (1976). The overall scale reliability was acceptable at 0.86. The six-item information seeking scale had an alpha reliability of 0.82. Technological innovativeness within both domains extracted in Study 1, was measured separately using Goldsmith and Hofacker's (1991) six-item Domain-Specific Innovativeness scale, with an alpha reliability of 0.86, and 0.81 respectively. Last, the alpha reliability for likelihood to purchase within the two dimensions was 0.82, and 0.82, respectively. Prior technology ownership, cosmopolitaness, and integrated social networks were each reduced to single composite indicators by adding their respective dummy coded items, while media use was valued by the mean of items measuring television viewing and Internet use.

Evaluation of Structural Model

Though the original data analysis examined technological innovativeness within two separate domains, the results were fairly similar across models. So in the interest of parsimony, only the result for one model is reported. In evaluating the structural model, the reasonableness of the hypothesized model was assessed first. Joreskog and Sorbom (1986) suggest four conditions to assess the reasonableness of a model. First, the standard errors of parameter estimates should not be large; second, the parameter estimates should not be highly correlated; third, all variances should be positive definite; and fourth, all absolute correlations should be less than one. The overall model reasonably met all these conditions.

The next step in model testing was to estimate the goodness of fit of the hypothesized research model. This is

typically done using a chi-square (χ^2) test; however, such tests are sensitive to sample sizes. Hence, Bentler and Bonett (1980) suggest using a χ^2/df ratio (*df*: degree of freedom) also called the relative chi-square as a more appropriate measure of model fit. This ratio should not exceed 5 for models with a good fit (Bentler, 1989). Using this criterion, the value of 2.32 ($p < .001$) indicates a good fit for the hypothesized research model ($\chi^2 = 90.48$, $df = 39$). Since there is dissatisfaction with the chi-square and related tests, many supplemental tests have been proposed (Holbert & Stephenson, 2002). Other tests include *NFI* (Normed Fit Index), *CFI* (Comparative Fit Index) and *RMSEA* (Root Mean Square Error of Approximation; Browne & Cudeck, 1993). Specifically, the closer *NFI* and *CFI* approach unity, the better the model fits the data, with an ideal cut-off value close to 0.95. On the other hand, *RMSEA* values less than .05 indicate a good fit, values as high as .08 represent a mediocre fit, and those greater than .10 indicate poor fit (Browne & Cudeck, 1993; MacCallum, Browne, & Sugawara, 1996). Though each metric has its own limitations, the three measures collectively provide a reasonable estimation of overall model fit. The hypothesized model had a *NFI* of .77, *CFI* of .85, and *RMSEA* of .07, which collectively indicate a poor fit.

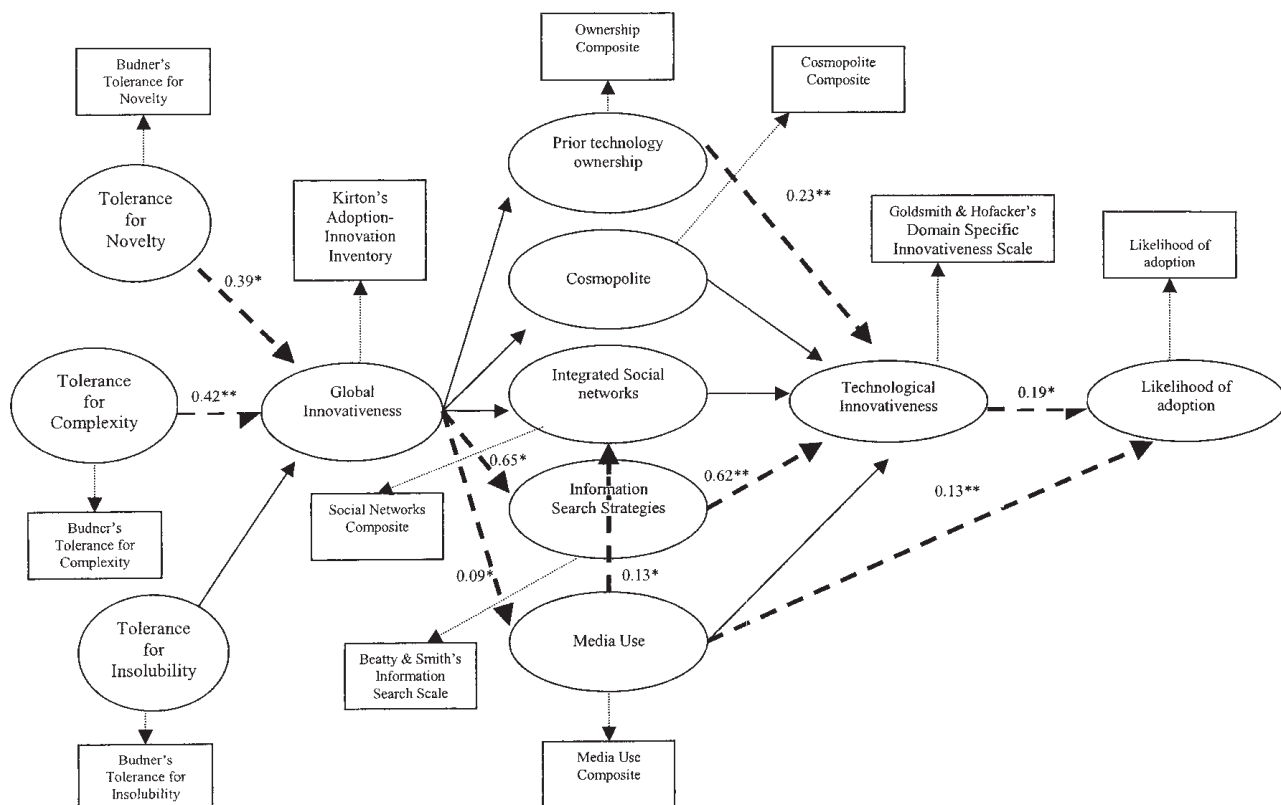
The next step in the evaluation process was to improve the fit of the model. Here the modification indices (*MI*) for the fixed and constrained parameters were scrutinized for misspecification (Joreskog & Sorbom, 1989). A modification index is a measure of predicted decreases in χ^2 if

any single constraint is added or removed and the model is re-estimated. Evidence of a relationship stood out for two associations, between media use and social networks, and between media use and the likelihood of adoption. Generally, any re-estimation of a model is warranted by the statistical parameters and the theoretical justification for the purported associations. The impact of television viewing on social networks has been suggested by Putnam (1996), and the impact of Internet use on the purchase of related technology has also been suggested in prior diffusion research (e.g., Vishwanath & Goldhaber, 2003).

Since the above associations met the required criteria, the model was re-estimated by constraining the three relationships. The reduction of 2 degrees of freedom resulted in a decrease in chi-square of 8.80. The overall fit for the revised model of $\chi^2 = 81.62$, $p < .01$, ($df = 37$); *NFI* = .83; *CFI* = .90; and *RMSEA* = .06 indicated a very good fit. The variances explained (R^2) for each of the endogenous variables were as follows: likelihood of adoption (0.16); technological innovativeness (0.45); prior technology ownership (.005); cosmopolite (.001); integrated social networks (.024); information seeking (.43); media use (.005); and global innovativeness (.33). The revised structural model is presented in Figure 2.

Hypotheses

Hypothesis 1 states that global innovativeness will be positively related to tolerance for ambiguity, complexity, and



Note: * $p < .05$ and ** $p < .01$. Global Innovativeness: $R^2 = 0.33$; Technological Innovativeness: $R^2 = 0.45$; Likelihood to Adopt: $R^2 = 0.16$.

FIG. 2. Revised structural model.

insolubility. In the revised structural mode, the links between tolerance for novelty ($\beta = 0.38, p < .05$), and tolerance for complexity ($\beta = 0.42, p < .01$) to global innovativeness were significant. Hence this hypothesis is partially supported.

Hypotheses 2 and 3 propose a direct impact of global innovativeness on a set of five communication-related behaviors, and a mediating role of global innovativeness on tolerance for uncertainty and these behaviors. In the revised model, only two variables, information search strategies ($\beta = 0.65, p < .05$) and media use ($\beta = 0.09, p < .05$) were significant. Hence these hypotheses are partially supported.

Hypothesis 4 predicts a direct impact of communication behaviors on technological innovativeness. The revised model found a significant impact for information search strategies ($\beta = 0.62, p < .01$), and prior ownership of technology ($\beta = 0.23, p < .01$). Hence this hypothesis was partially supported.

Last, Hypothesis 5 states that technological innovativeness will mediate the impact of communication behavior, and predict the likelihood to adopt new technology. In the revised model, technological innovativeness significantly predicted the likelihood of adoption ($\beta = 0.19, p < .05, R^2 = 0.16$). Technological innovativeness partially mediated the impact of communication and media use since there was a significant direct impact of media use on the likelihood of adoption ($\beta = 0.13, p < .01$).

Discussion

The research model integrates the influence of global innovativeness and context-specific innovativeness within the domain of technology adoption. It was hypothesized that global innovativeness would be impacted by the individuals' tolerance for uncertainty, while technological innovativeness would be conditioned by exposure to media and other communication behavior. These would indirectly impact the likelihood of technological adoption. The research model was tested over two technology clusters, and the results were fairly consistent.

The first set of hypotheses examines the antecedents of global innovativeness. Though global innovativeness is considered a trait, Rogers and Shoemaker (1971) argued that since variables such as educational attainment and occupational status correlated with global innovativeness, it is more plausible that this trait is not a genetic constant, but also socially influenced. Hence, the research model controlled for external variables such as education and occupation by choosing a student sample, and hypothesized an impact of tolerance for uncertainty on global innovativeness. In testing the impact of an individual's tolerance for uncertainty, it was found that tolerance for novelty, and tolerance for complexity significantly impacted global innovativeness. This finding is important and points to the importance of tolerance for uncertainty as a scope condition in partially defining an individual's global innovativeness. Most new innovations have few users, and limited experiential cues for a potential innovator. In addition, new innovations often entail complexity,

which translates to a need to learn to use an innovation. This would require some degree of effort towards understanding it uses and applications. To cope with this, any innovator would require a supporting personality that tolerates high levels of novelty, and complexity in a given context.

The relationship between tolerance for novelty, global innovativeness, media use, and information seeking is consistent with prior theoretical propositions by Hirschman (1980). Tolerance for novelty can be considered as an indicator of novelty seeking behavior. Novelty seeking behavior is a global personality trait, which represents an innate search for information. Actualized novelty seeking refers to the initiation of behaviors intended to acquire new information, and is conceptually distinct from global innovativeness (Hirschman, 1980). This information functions as a means of self-preservation, improves problem solving, and often leads to the actual adoption of innovations (Hirschman, 1980). Hence novelty seekers are attracted to information-rich media that absorb, digest, and chronicle multiple sources and multiple topics of information. The mediated relationship of global innovativeness supported by the research model presents empirical validation to Hirschman's notion of novelty seeking and innovativeness, and suggests an additional role of tolerance for complexity in predicting global innovativeness.

Next, global innovativeness was hypothesized to impact a host of communication behaviors. Based on Rogers (1995), it was hypothesized that innovators would possess certain characteristics because of their personality traits. Since the study focused on technological innovativeness, it was hypothesized that global innovativeness would result in a high ownership of technology products. Being obsessively venturesome would also result in innovators being open-minded and cosmopolite, with more links outside the boundaries of their social system. In addition, they would have integrated and interconnected interpersonal social networks that include other innovators. Global innovators would also be more adept at searching for information about various technological products. In addition, they would tend to use new media such as the Internet, and other mass media. In testing these links, support was found for two propositions: global innovators tended to be better at information searching, and tended to have higher media use. The variables tested for information searching probed the likelihood to ask questions to the salesperson prior to purchase, asking others about the product, seeking information through other sources, comprehensively researching a product prior to use, and trying out the product prior to use. Significance of this variable translates to sophistication in information search. Global innovators with their higher tolerance for novelty and complexity are more adept at probing other sources for information prior to product purchase. They tend to be willing to experience and try different modes of information, and are actively seeking this information. The significant relationship between media use and global innovativeness furthers this point. Since new media such as the Internet serve as an immediate compendium of vast amounts of information,

innovators tend to use this medium to learn about and research new products. Their penchant for venturesomeness, supported by their higher tolerance for novelty and complexity then influences their interest and likelihood of purchasing new technology products. Further support for this hypothesis is suggested by the significant relationship between media use and likelihood of technology purchase.

Next, the research model presents an information processing view on attitude formation, and hypothesized that an individual's proclivity to innovate within any specific context was conditioned by his or her exposure, experience, and communication behaviors. Within the context of technological innovativeness, it was hypothesized that prior ownership and use of technology would create a willingness in the minds of individuals to adopt that technology, especially if they were already possessing a personality that supported innovativeness. It was also purported that exposure to other innovators and their experiences would contribute to innovativeness within a specific domain. That is, if individuals have friends who tend to be technological innovators themselves and who possess a number of technology products, these friends would, in turn, influence others to adopt. However, to have social networks of innovators, individuals would require a personality that accepts innovativeness and tolerates uncertainty. In addition, sophistication in information search strategies and exposure to media such as television and the Internet could also influence an individual's interest in a specific context, and thereby his or her domain-specific innovativeness. In testing these hypotheses, the relationship between information search strategies and prior technology ownership on domain-specific innovativeness were significant. Prior ownership translates to interest and experience within a particular domain. Coupled with sophistication in information searching, this translates to an interest that is furthered by actively searching for information.

Last, technological innovativeness along with media use significantly predicted the likelihood of technology adoption. The ability to significantly predict a behavior is a powerful indicator of the validity of a model, and these findings lend credence to it. Hence, likelihood to adopt a technology is predicted by an individual's degree of technological innovativeness and his or her exposure to mass media. Technological innovativeness is determined by experience with and interest in other technology, and by the person's sophistication in searching for relevant information. Information search strategies are determined by the person's innate innovativeness, which, in turn, is determined by the tolerance for novelty, and complexity.

The research model tested in the study has some noteworthy limitations. The research model uses only tolerance for uncertainty as a scope condition determining global innovativeness. In addition to tolerance for uncertainty, Roger's (1995) suggests opinion leadership, degree of dogmatism and interest in science as indicators of innovativeness. However, the measures used for global innovativeness, and domain-specific innovativeness tend to correlate with opinion leadership. Also, measures for opinion leadership

are almost identical to those measuring tolerance for uncertainty. Hence, there seems to be a high likelihood of multicollinearity. So, in the interest of parsimony, the focus of the model was on only one scope condition. Next, the reliability of a model is limited to the measures used and the theoretical justifications for their use. However, variables, no matter how reliable or valid, are limited in their ability to measure a construct. Since all models are limited by the number of variables they include, the impact of variables not included in a model cannot be ascertained. Next, since it is difficult to determine the boundaries of technology products, the research used a cluster of products, generated *a priori*, for which the likelihood of adoption was elicited. Each cluster consisted of six to seven functionally similar products. However, each product within a cluster could be considered a separate domain, and could potentially influence the results of the study. Also, the use of student samples has inherent limitations. It is often assumed that using student samples can significantly control for variations in income and aspiration (some of the factors suggested by Rogers), which may not always be the case. Last, the study measured all the variables using one instrument at a single point in time. This can result in method variance, which could impact the reliability of the results. Hence, future research needs to replicate and validate these findings, and determine the robustness of these results by extensions to other domains. Since most of the products included in this study constituted new technological innovations, future research could test this model against a list of late majority products or with mature products to check the reliability of the model. The addition of other personality factors could also enhance the ability of model to predict potential purchase.¹

In conclusion, the model presents an integration of the impact of global behavior, and context-driven behavior on technological innovativeness, and finds antecedents for both. In doing so, it suggests an inclusive treatment of personality factors when predicting innovativeness. The results of the study also point to a prominent role of sophistication in information search in predicting innovativeness. Incorporating these factors in future communication research should lead to a better understanding of the diffusion phenomenon.

¹The research model presented is one interpretation of the relationship between innate behavior and context-specific behavior. It is based on theoretical and empirical reasoning suggested by prior research. Another interpretation could be that innate behavior might not mediate communication behavior, but rather influence context-specific behavior along with other communication and media use variables. Empirical evidence suggests the contrary. Foxall and Szmigin (1999) compared Goldsmith and Hofacker's (1991) Domain-Specific Innovativeness measure and Kirton's Innovativeness Inventory (KAI) and found no significant correlation between the two. In any case, to validate these results and test the assumptions about the role of innate innovativeness, a structural model was created where innate innovativeness did not mediate communication behaviors, but impacted context-specific innovativeness along with all the other communication variables. The results confirmed the lack of direct relationship between innate innovativeness and domain-specific innovativeness, and resulted in a very poor model fit, χ^2/df ratio = 2.46, $p < .01$, ($df = 39$), $NFI = 0.68$, $CFI = 0.71$, $RMSEA = 0.08$.

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